

The range of visible spectrum is 430nm (violet) to about 790nm (red)

- ② An image may be continuous w.r.t. to the x and y -coordinates and also in amplitude. To digitize it, we have to sample the function in both coordinates and also in amplitude. Digitizing the coordinate values is called sampling and digitizing the amplitude values is called quantization.

$f(s, t)$ represents a continuous image function of two continuous variables s and t . We convert this function into a digital image by sampling and quantization.

We sample the continuous image into a digital image $f(x, y)$, containing M rows and N columns where (x, y) are discrete coordinates $[x = 0, 1, 2, \dots, M-1, y = 0, 1, 2, \dots, N-1]$

The section of the real plane spanned by the coordinates of an image is called spatial domain, with x and y being referred to as spatial variables or spatial coordinates.

In grayscale, (x, y, z) $(x, y) \rightarrow$ spatial coordinates, z is the value of f at coordinates (x, y)

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \dots & f(0, N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1, 0) & f(M-1, 1) & \dots & f(M-1, N-1) \end{bmatrix}$$

Each element of this array is called an image element, picture element, pixel or p.e.

$$A = \begin{bmatrix} a_{0,0} & a_{0,1} & \dots & a_{0,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \dots & a_{M-1,N-1} \end{bmatrix} \quad \text{We use } f(i, j) \text{ when referring to a pixel with coordinates } (i, j)$$

$a_{ij} = f(i, j)$

Euclidean distance, $D_e(P, Q) = \left[(x-u)^2 + (y-v)^2 \right]^{\frac{1}{2}}$

City block distance, $D_4(P, Q) = |x-u| + |y-v|$

Chessboard distance, $D_8(P, Q) = \max \{ |x-u|, |y-v| \}$

The centre of an $M \times N$ digital image with origin at $(0,0)$ and range to $(M-1, N-1) \Rightarrow (x_c, y_c) = (\text{floor}(M/2) + 1, \text{floor}(N/2) + 1)$

The sampling process may be viewed as partitioning the xy -plane into a grid, with the coordinates of the centers of each cell in the grid being a pair of elements from the Cartesian product $\mathbb{Z}^2$ [set of all ordered pairs of elements (z_i, z_j)]

Hence, $f(x, y)$ is a digital image if (x, y) are integers from $\mathbb{Z}^2$ and f is a funcⁿ that assigns an intensity value ($i \in \mathbb{R}$) to each distinct pair of coordinates (x, y) . This functional assignment is the quantization process.

If the intensity levels are also integers, then $R = \mathbb{Z}$ and a digital image becomes a 2D-funcⁿ whose coordinates and amplitude values are integers.

Restriction $M, N > 0$ $L = 2^k$ [k is integer]
We assume that the discrete levels are equally spaced and they are integers in the range $[0, L-1]$

Dynamic Range = The ratio of the maximum measurable intensity to the minimum detectable intensity level in the system.

The upper limit is determined by the saturation and lower limit by noise.

Image Contrast = Difference in intensity ~~levels~~ between the highest and lowest intensity levels in an image.
Contrast Ratio is the ratio of highest to lowest.

The no. of bits required to store a digital image is, $b = M \times N \times k$ → discrete random variable

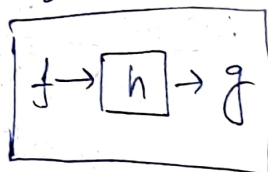
when $M = N$, then, $b = N^2 k$

Image can have 2^k possible intensity levels $\Rightarrow$ "k-bit image"

Convolution of two functions $f(x)$ and $h(x)$

$$g(x) = f(x) * h(x) = \int_{-\infty}^{\infty} f(\tau) h(x-\tau) d\tau$$

Convolution is Linear Shift Invariant System.



Linearity,

$$\text{let } g_1(x) = \int_{-\infty}^{\infty} f_1(\tau) h(x-\tau) d\tau$$

$$g_2(x) = \int_{-\infty}^{\infty} f_2(\tau) h(x-\tau) d\tau$$

Then,

$$\int_{-\infty}^{\infty} (\alpha f_1(\tau) + \beta f_2(\tau)) h(x-\tau) d\tau = \alpha g_1(x) + \beta g_2(x)$$

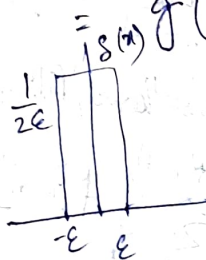
Shift Invariance,

$$\text{let, } g(x) = \int_{-\infty}^{\infty} f(\tau) h(x-\tau) d\tau$$

$$\text{Then, } \int_{-\infty}^{\infty} f(\tau-a) h(x-\tau) d\tau = \int_{-\infty}^{\infty} f(\mu) h(x-a-\mu) d\mu \quad [\mu = \tau-a]$$

Unit Impulse function,

$$\delta(x) = \begin{cases} 1/2\epsilon & |x| \leq \epsilon \\ 0 & |x| > \epsilon \end{cases} \quad \epsilon \rightarrow 0$$



$$\int_{-\infty}^{\infty} \delta(\tau) d\tau = \frac{1}{2\epsilon} \cdot 2\epsilon = 1$$

$$\Rightarrow \boxed{\int_{-\infty}^{\infty} \delta(\tau) b(x-\tau) d\tau = b(x)} \rightarrow \text{shifting property}$$

$$h(x) = \delta(x) * h(x)$$

$$h(x) = \int_{-\infty}^{\infty} \delta(x) h(x-z) dz$$

distant star $\delta(x, y) \rightarrow$ Human Eye $\rightarrow$ Point Spread Function $h(x, y)$

Properties of Convolution:-

Commutative: $a * b = b * a$

Associative $(a * b) * c = a * (b * c)$

$$f \rightarrow [h_1] \rightarrow [h_2] \rightarrow g \equiv f \rightarrow [h_1 * h_2] \rightarrow g$$

2D convolution:-

$$2DSS: f(x, y) \rightarrow [h(x, y)] \rightarrow g(x, y)$$

convolution:

$$g(x, y) = \iint_{-\infty}^{\infty} f(x, y) h(x-x, y-y) dx dy$$

Convolution with Discrete Images:-

$$f[i, j] \rightarrow [h[i, j]] \rightarrow g[i, j]$$

$\Downarrow$
Mask/Kernel/Filter

$$g[i, j] = \sum_{m=1}^M \sum_{n=1}^N f[m, n] h[i-m, j-n]$$

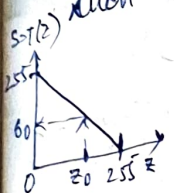
SPATIAL OPERATIONS:-

Spatial operations are performed directly on the pixels of an image.

(i) Single Pixel operations:-

The simplest operation we perform on a digital image is to alter the intensity of its pixels individually using a transformation function, T , such that

$$s = T(z) \quad \begin{cases} z = \text{intensity of a pixel in original image} \\ s = \text{intensity of that pixel in processed image} \end{cases}$$



$$T(f(x, y))$$

Ex: $F(x, y) = 255 - F(x, y) \Rightarrow$ to obtain the negative image.

(ii) Neighbourhood Operations:-

The value of the pixel is determined by a specified operation on the neighbourhood of pixels in the input image.

$$g(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} f(r, c)$$

for 8-neighbourhood, (3×3)

$$g(x, y) = \frac{1}{9} \sum_{i=x-1}^{i=x+1} \sum_{j=y-1}^{j=y+1} (f(i, j))$$

(iii) Geometric Transformations:-

We use geometric transformations to modify the spatial arrangement of pixels in an image. These transformations are called "rubber sheet transformations" because they may be viewed as analogous to "printing" an image on a rubber sheet, then stretching or shrinking the sheet according to a predefined set of rules. It consists of two basic operations:

(a) Spatial transformation of coordinates

(b) Intensity interpolation that assigns intensity values to the spatially transformed pixels. (Interpolation is the process of using known data to estimate values at unknown location.)

Affine transformations (scaling, translation, rotation, shearing)

The key characteristic of an affine transformation in 2D is that it preserves points, straight lines and planes.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = A \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

The preceding transform moves the coordinate pixels in an image to new locations. To complete the process we have to assign

In homogeneous coordinates we can perform all affine transformations and also concat a set of transformations.

Rotation in anticlockwise direction

$$\begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Scaling

$$\begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$S_x = -1, S_y = -1 \Rightarrow$ reflection about X
 $S_x = -1, S_y = 1 \Rightarrow$ " " Y

Shear

$$\begin{bmatrix} 1 & Sh_x & 0 \\ Sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

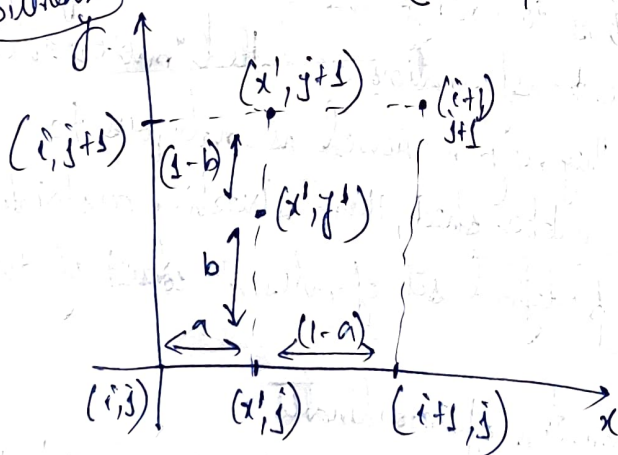
Translation

$$\begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix}$$

Intensity Interpolation

(Affine Transformation for mapping)
 Interpolation for intensity

Bilinear



$$i \leq x' \leq i+1$$

$$j \leq y' \leq j+1$$

$$x' = i + a$$

$$y' = j + b$$

Interpolating in x -direction $\rightarrow f'(x', j) = (1-a)f(i, j) + af(i+1, j)$

Interpolating in y -direction $\rightarrow f(x', y') = (1-b)f(x', j) + bf(x', j+1)$

③ ut

$$f'(x', j+1) = a f'(i+1, j+1) + (1-a) f'(i, j+1)$$

limit the

for nearest neighbour,

original $\leftarrow$ spatial
 $(x, y) \quad (x', y')$

$$f'(x', y') = F \left(\lfloor x+0.5 \rfloor, \lfloor y+0.5 \rfloor \right)$$

Image Transforms:- (Transform Domain Operations)

spatial domain $\rightarrow f(x, y)$ [Input image]

2D - Fourier transform

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi xy} e^{-j2\pi yv}$$

$e^{-j2\pi xy} e^{-j2\pi yv}$ is called forward transformation kernel

$\begin{bmatrix} u = 0, 1, 2, \dots, M-1 \\ v = 0, 1, 2, \dots, N-1 \end{bmatrix}$ (x, y) are spatial variables
 u, v are transform variables.

$T(u, v)$ is called the forward transform of $f(x, y)$

Inverse Transformation of $T(u, v)$:

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) s(x, y, u, v)$$

[For, $\begin{bmatrix} x = 0, 1, 2, \dots, M-1 \\ y = 0, 1, 2, \dots, N-1 \end{bmatrix}$]

$s(x, y, u, v)$ is called an inversed transformation kernel.

① The forward transformation kernel is said to be separable if

$$e^{-j2\pi xy} e^{-j2\pi yv} = e^{-j2\pi xu} \cdot e^{-j2\pi yv}$$

② The kernel is said to be symmetric if $e^{-j2\pi xu}$ is functionally equal to $e^{-j2\pi yv}$ so that

$$e^{-j2\pi xy} e^{-j2\pi yv} = e^{-j2\pi xu} \cdot e^{-j2\pi yv}$$

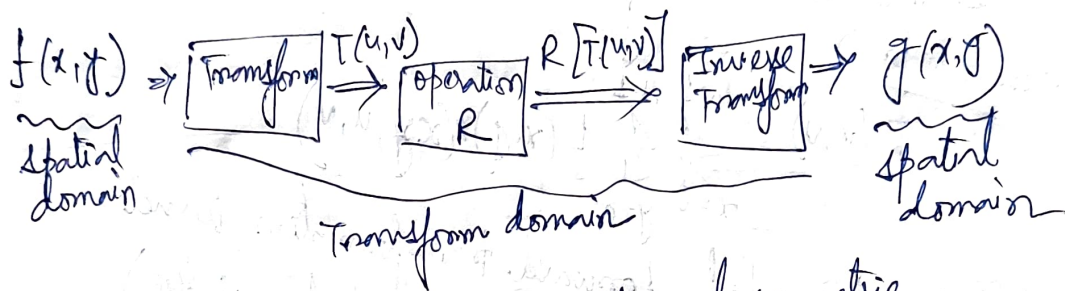
i.e. $e^{-j2\pi yv} = e^{-j2\pi xu}$

The nature of transform is determined by its kernel,
A transform of particular importance in image processing
is Fourier Transform.

$$r(x, y, u, v) = e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)} \quad [j = \sqrt{-1}]$$

and

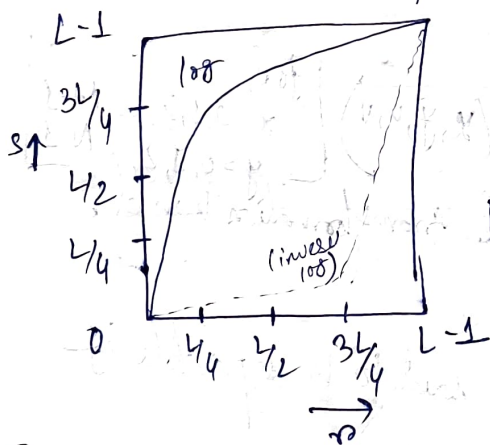
$$s(x, y, u, v) = \frac{1}{MN} e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$



Fourier kernels are also separable and symmetric

Log Transformation $\rightarrow s = c \log(s+r)$ $r \geq 0$

$\uparrow$ output $\uparrow$ constant $\uparrow$ Input



We use this kind of transformation
to expand the values of dark
pixels in an image, while compressing
the higher-level values.

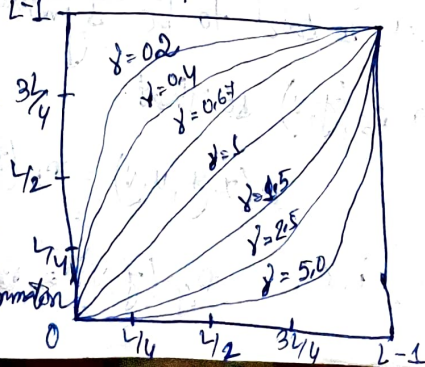
⑧ Power Law / Gamma Transformation / Correction

$$s = c \cdot r^\gamma \quad [c \text{ and } \gamma \text{ are positive constants}]$$

If $\gamma < 1$ logarithmic / n th root

If $\gamma > 1$ exponential / n th power

when $c=1, \gamma=1$ = Identity transformation



The response of many devices used for image capture, printing and display obey a power law. By convention, the exponent in a power-law equation is referred to as gamma. The process used to correct these power-law response phenomena is called gamma correction/encoding. Ext-CRT devices have an intensity-to-voltage response that is a power function, with exponents varying from approximately 1.8 to 2.5.

As with log-transformations, power-law curves with fractional values of γ map a narrow range of dark input values into a wider range of output values, with the opposite being true for higher values of input levels. A family of transformations can be obtained by varying γ .

